

VLSI DESIGN
EE-671
ASSIGNMENT-2

Note:

1. Plots and Graphs

- Ensure that all plots included in your report are **clearly legible and readable**.
- Each axis must be labelled with appropriate **values and units**.

2. Tools and PDK Requirement

- Install and use **IIC-OSIC-TOOLS**.
- Use only the **Sky130A PDK** for this assignment.

3. Submission Format

- Submit your work as a **ZIP file** containing the following:
 - All code files used for the simulations with layout files.
 - A **Report in PDF format** including:
 - Code snippets.
 - Graphs/plots with results (**screenshots**).
 - Explanations and observations.
 - Layout with measurement (**screenshots**)

4. Simulation Parameters

- For all simulations, use a supply voltage of **VDD = 1.8 V**.
-

Introduction:

In this assignment you will (a) layout the standard cells designed in assignment-1 that are DRC (Design rule check) clean, (b) perform LVS (layout versus schematic) check, (c) perform a parasitic extraction (PEX) from the layout and (d) simulate the PEX netlist.

Tools:

- (a) You will use **Magic** (an open-source tool) for layout, netlist extraction (with and without parasitics).
 - (b) **Netgen** (an open-source tool) for LVS (Layout-vs-schematic). An LVS check ensures that the layout and the original circuit/schematic/Netlist represent the same thing.
 - (c) **NGSpice** (a spice circuit simulator) tool to simulate the circuit/netlist.
-

Q1)

a) For the INVX1 designed in assignment-1, draw the layout using Magic tool.

The layout tool will open and you can start drawing the layout. The standard cell layout that you create **must** have a height of 2.72 μm and must be DRC clean. An example screenshot of an INVX1 is shown in Fig.1 for your reference.

Please refer to the tutorial links (other links have already been provided)

[Creating a Hierarchical Layout in Magic Using the Sky130 PDK](#)

[Magic VLSI Tutorial \(part 2\), Laying out CMOS inverter](#)

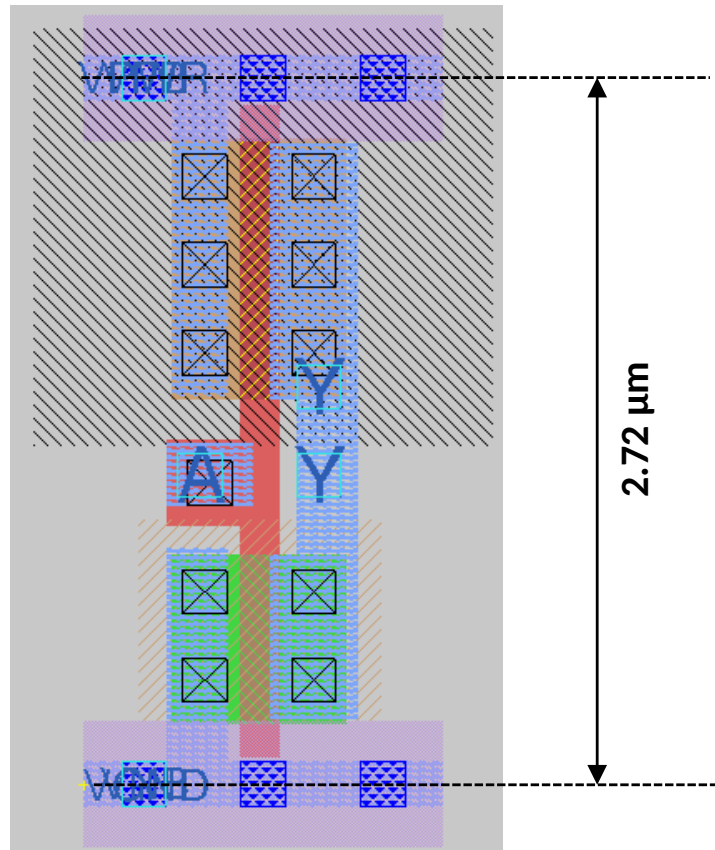


Figure 1: Example layout of INVX1 with height 2.72 um (mid-point of metal-1 VDD to mid-point of metal-1 VSS line)

In the tool, you must see  indication, indicating there are zero DRC errors.

b) We will now extract the netlist (for LVS) from this layout using Magic. In the **Magic console**, type the following commands:

```
% extract all
% ext2spice lvs
% ext2spice -d -o invx1_layout.spice
```

You will now have a new netlist file "invx1_layout.spice", extracted from the netlist. This netlist needs to be compared with the netlist that you wrote for the INVX1 design in assignment-1. In the main terminal use **netgen** to run the lvs using the following command:

```
netgen -batch lvs "invx1_layout.spice INVX1" "invx1_schematic.cir INVX1" /foss/pdks/sky130A/libs.tech/netgen/sky130A_setup.tcl
```

(We can save the ngspice schematic file using either the **.cir** or **.spice** extension.)

In this command, you need to point to the two netlists (_layout.spice and _schematic.cir) and the subckt names in the netlist (INVX1 in my case). If the LVS is clean, you must obtain the following:

```
Circuit 1 contains 2 devices, Circuit 2 contains 2 devices.
Circuit 1 contains 6 nets, Circuit 2 contains 6 nets.

Final result:
Circuits match uniquely.
Logging to file "comp.out" disabled
LVS Done.
```

c) We will now extract another netlist from the layout, but with all parasitic information, for our simulations. In the **Magic console**, type the following commands:

```
% extract all
% ext2spice scale off
% ext2spice cthresh 0
% ext2spice rthresh 0
% ext2spice -d -o invx1_pex.spice -f ngspice
```

You will now have a new netlist file “invx1_pex.spice”, extracted from the netlist with all parasitic information. Open this file and notice the additional parasitic caps at different nodes.

You can now use this netlist to perform simulations similar to what you did in assignment-1 (replace the original netlist with this new netlist extracted from layout).

In your report, you need to show the following:

1. Screenshot of your layout with the height of the layout annotated.
2. DRC clean screenshot.
3. LVS clean screenshot.
4. The following table:

Inverter Dynamic Characteristic	Schematic	Layout
Rise time, t_r (ps)		
Fall time, t_f (ps)		
Propagation delay, t_p (ps)		

Q2) For the strength-2 inverter (**INVX2**) designed in assignment-1, repeat the Q1 above.